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Object-hanging device

Abstract

Object-hanging device (**1**), intended to couple reversibly with a support wall (**2**) with a vertical extension and equipped with suitable through holes (**3**), comprising a support body (**10**), intended to support in suspension an external object and a fixing body (**100**), adapted to reversibly engage with the support body (**10**) to couple the support body (**10**) reversibly with the wall (**2**).

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3715096	12/1972	Filbert	248/220.22	A47F 5/0823
4258892	12/1980	Craine	N/A	N/A
4289242	12/1980	Kenyon	211/59.1	A47F 5/0861
5180128	12/1992	Massey	D8/367	A47F 5/0823
10827853	12/2019	Ma	N/A	E05B 65/00
2003/0230018	12/2002	Poulokefalos	40/642.02	A47F 5/0823
2005/0029205	12/2004	Mansfield	211/7	A47F 5/0861
2009/0049723	12/2008	Arndt	40/642.01	A47F 5/0823
2009/0173853	12/2008	Fawcett	248/220.31	A47F 5/0861
2019/0290023	12/2018	Ma	N/A	A47F 5/0861

OTHER PUBLICATIONS

Italian Search Report, issued in IT 2022/00026955 on May 3, 2023, 8 pages. cited by applicant

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Background/Summary

(1) The object of the present invention is an object-hanging device intended to be coupled reversibly with a support wall with a vertical extension and equipped with suitable through holes.
(2) A support wall means an object designed to exhibit objects on walls or on other vertical surfaces.
(3) For example, the wall can be employed for use in workshops or warehouses or other work areas in which work tools are required that are arranged visibly so as to be able to be exploited rapidly when required.
(4) Alternatively, this wall can be used as an element to exhibit objects or products to the public. For example, the wall can be installed in a shop window or inside premises used for commercial activities or in trade fairs. Using object-hanging devices arranged on drilled support walls is rather common in the prior art. Such walls generally have a flat shape and comprise a first side and a second side. These walls are provided with a plurality of holes, arranged respectively at regular intervals both horizontally and vertically, forming a matrix of holes. These holes are generally square or rectangular in shape and pass through the thickness of the wall.
(5) Generally, this wall has a frontal surface, exposed towards the observer, and a rear surface, arranged opposite the wall relative to the frontal surface and an object-hanging device is arranged

on the frontal surface of the wall. A common embodiment of this device comprises suitable rear hooking teeth that are engaged inside the holes of the wall as supports, or in other words hooking onto these holes of the wall so as to maintain the objects hanging. Disadvantageously, this type of device is not fixed and has a great risk of unhooking from the support wall in the event of an accidental collision with an operator, especially if this collision comes from below.
(6) In order to overcome this drawback, there are different systems in the prior art for improving fixing such devices to the support walls.
(7) The most common system is using auxiliary fixing systems using screwing members. Nevertheless, these systems, in addition to being costly, are laborious to install and remove because they need equipment for screwing the members tight. This aspect is particularly disadvantageous during exhibiting at trade fairs where dressing the wall and relative objects being exhibited is required within a short time but at the same time solid and secure fixing of the objects to the wall is required.
(8) Alternative fixing system can include devices using pressure-locking systems where the systems work by means of mechanical interference or friction. In particular, these systems comprise connecting members that are insertable into the holes and compatible in shape therewith.
(9) Disadvantageously, these systems on the other hand do not ensure a sufficiently secure lock of the device on the wall, in particular in the case of vibrations, which would tend to unhook the device from the wall.
(10) The technical task of the present invention is to provide an object-hanging device intended for being coupled reversibly with a support wall that enables the drawbacks that have emerged from the prior art to be overcome.
(11) One object of the present invention is accordingly to provide an object-hanging device that is easily installable and uninstallable in a reversible manner.
(12) A further object of the present invention is to provide an object-hanging device that does not rotate or unhook if the wall is stressed by impacts or vibrations, and in particular, that prevents unhooking from this support wall in the event of an impact from below arising from an accidental collision with an operator.
(13) A further object of the present invention is to provide an object-hanging device that is cheap, easily makeable and reusable.
(14) The technical task set and the objects specified are substantially attained by an object-hanging device according to the technical features as set out in one or more of the accompanying claims.
(15) In particular, the technical task set and the objects specified are substantially achieved by an object-hanging device comprising a support body and a fixing body.
(16) The fixing body comprises a first portion configured to make a rest and/or a stop for an object and a second portion configured to make a reversible connection between the support body and the support wall.
(17) The fixing body is reversibly engageable or engaged with the first portion and/or with the second portion in a use configuration of the object-hanging device.
(18) According to one aspect of the present invention, the fixing body is reversibly interposed between the first portion and the second portion in the use configuration of the object-hanging device.
(19) In the use configuration, the second portion is at least partially inserted inside at least one of the holes so as to define an anchoring of the support body on the wall.
(20) In the use configuration, the fixing body exerts a first thrust action and a second thrust action, opposite the first thrust action, respectively acting on the first portion, and/or on the second portion, and on the wall so as to define stable anchoring of the support body on the wall.
(21) According to one aspect of the present invention, the first portion preferably has a longitudinal dimension that is more extended than the other dimensions. The longitudinal geometry of the first portion can be respectively at least partially linear or broken or curved. The first portion has two

ends, a first end of which is engaged with an external object whereas a second end is arranged near or adhering to the wall and can comprise a geometry that is such as to form an auxiliary support rest for the wall.
(22) According to one aspect of the present invention, the first portion extends along its longitudinal axis, having a first end and a second end provided with respective transverse elements, said transverse elements being opposite one another relative to said longitudinal axis: a first transverse element is arranged along the first end and engages with the external object and a second transverse end is adapted to provide a rest against the frontal surface of the wall.
(23) According to one aspect of the present invention, the fixing body engages the support body along the frontal surface of the wall, engaging respectively with the support body, with the first section and/or with the second section, anchoring the support body to the frontal surface of the wall.
(24) According to one aspect of the present invention, the fixing body comprises at least one first seat suitable for receiving a longitudinal portion of said first portion of the support body. This seat can be so arranged as to engage with said support body arranged outside or inside the fixing body.
(25) According to a further aspect of the present invention, the fixing body is so configured as to prevent the support body from rotating around any axis perpendicular to the wall.
(26) According to one aspect of the present invention, the support body can further comprise a third portion, defined between said first portion and said second portion, defining a housing for or an element for coupling with said fixing body.
(27) According to a different aspect of the present invention, the support body has an overall belt-shaped geometry. The second portion is inserted into a through hole on the support wall and has a linear rest on the rear surface of the support wall. The first portion, with a substantially linear geometry, is arranged on the frontal surface of the wall. The fixing body is so configured as to receive internally in a suitable hole the first portion and to retain by means of an internal gripping member.
(28) Advantageously, using a device comprising a support body and a fixing body that is reversibly engageable with the support body permits a reversible coupling with the support wall, anyway ensuring that the device is stably anchored on the wall and is not subjected to involuntary movements, whether in the event of impacts or of vibrations.
(29) Advantageously, the above device is easy to install and uninstall.
(30) Further, the device is realizable easily and economically.
(31) Further, the device is easily reusable to support objects.
(32) Additional features and advantages of the present invention will become more apparent from the indicative and thus non-limiting description of a preferred but not exclusive embodiment of an object-hanging device intended to couple reversibly to a support wall with a vertical extension and equipped with suitable through holes.

Description

(1) Such a description will be set out below with reference to the accompanying drawings, which are provided solely for illustrative and therefore non-limiting purposes, in which:
(2) FIG. **1** is a frontal view of an object-hanging device coupled with a support wall provided with suitable holes that is the subject-matter of the present invention;
(3) FIG. **2** shows a side view of the device of FIG. **1**;
(4) FIGS. **3**, **4**, **5**, **6**, **7**, **8**, **9** and **10** are views of the consecutive passages of the coupling and decoupling of the device shown in FIG. **1** with and from the support wall;
(5) FIG. **11** is a view illustrating in a detailed frontal view the main component of the fixing body comprised in the device shown in FIG. **1**.

(6) FIG. **12** is a view illustrating an oblique top view of the main component of the fixing body shown in FIG. **11**.
(7) FIG. **13** is a view illustrating an oblique bottom view of the main component of the fixing body shown in FIGS. **11** and **12**;
(8) FIG. **14** is a view illustrating in a detailed bottom view the main component of the fixing body comprised in the device shown in FIG. **1**;
(9) FIG. **15** is a view illustrating an oblique top view of the support component of the fixing body shown in FIG. **14**.
(10) With reference to the appended figures, **1** denotes an object-hanging device intended to be coupled reversibly with a support wall **2** with a vertical extension and equipped with suitable through holes **3**.
(11) The support wall **2** can be of any type present in the prior art. The wall **2** shows a frontal surface **2***a* and a rear surface **2***b*.
(12) The term “frontal” means the surface arranged opposite an observer, i.e. directly in view relative to the point of view of the observer.
(13) The wall **2** can have a thickness of different values, but this thickness is preferably constant along the entire body of the wall **2**. The holes **3** can have different geometries and dimensions, preferably with a square or rectangular geometry and dimensions compatible with other fixing devices present in the prior art, having still more preferably a horizontal and/or vertical dimension comprised between 5 mm and 50 mm. The holes **3** are preferably arranged reciprocally vertically and/or horizontally at fixed distances, constituting regular vertical and/or horizontal series that therefore define a matrix of holes that is distributed regularly on the wall **2**.
(14) The device **1** is intended to support an external object (not shown) suspended along the wall **2**, in particular exerting a constraining reaction opposing the weight force “P” of the object.
(15) The device **1** comprises a support body **10** and a fixing body **100**.
(16) The support body **10** and the fixing body **100** can consist of different, preferably rigid, materials with great mechanical resistance to stress.
(17) Coupling with the wall can comprise a direct contact with the wall **2** or by means of coatings affecting the contact zones of the fixing body **100** or of the support wall **10** with the wall, which are preferably rough and/or rubberized, to improve the stability of the contact with the wall **2**.
(18) The support body **10** comprises a first portion **11** configured to make a rest and/or a stop for the external object to be hung along the wall **2** and a second portion configured to make a reversible connection between the support body **10** and the support wall **2**.
(19) In particular, the first portion **11** can comprise sections of different geometry, selected according to the type of object to be supported. Advantageously, the geometry of the first portion **11** can allow great flexibility of use of the object-hanging device **1**, which can accordingly support objects of different shapes and/or dimensions.
(20) The first portion **11** can be arranged cantilevered on the frontal surface **2***a* of the wall **2**, perpendicularly or askew relative to the frontal surface **2***a* of the wall **2** in function of the type of external object to be supported.
(21) Preferably, the first portion **11** has a geometry that is more developed along the longitudinal dimension than over the other dimensions.
(22) The longitudinal geometry can be respectively at least partially linear or broken or curved.
(23) The first portion **11** has two ends, a first end **11***a* of which is preferably free and dedicated to engaging with the external object whilst a second end **11***b* is arranged near or adhering to the frontal surface **2***a* of the wall **2**.
(24) According to the embodiment illustrated into the present figures, the first portion **11** extends along a longitudinal axis “S” thereof, in a substantially linear manner, and has a first end **11***a* and a second end **11***b* preferably provided with respective transverse elements. The transverse elements are opposite one another relative to the longitudinal axis “S”.

(25) Preferably, a first transverse element **11***c* is arranged along the first end **11***a* and is dedicated to engaging with the external object and a second transverse element **11***d* is arranged along the second end **11***b* and is adapted to form an auxiliary support rest **6** against the frontal surface **2***a* of the wall **2**. The presence of the transverse elements **11***c* and **11***d* is optional; nevertheless, the presence of the second transverse element **11***d* and the support rest **6** deriving therefrom may be necessary if the weight of the body to be maintained suspended may cause involuntary bending of the support body **10**.
(26) The second portion **12** is at least partially inserted between at least one of the through holes **3** so as to define anchoring of the support body **10** to the wall **2**. When the device **1** is in a use configuration, the second portion **12** engages with the rear surface **2***b* of the wall **2**, having at least one rest **4** on the rear surface **2***b*. In particular, the second portion **12** has a first sub-portion **12***a* intended to remain at least partially inside the hole **3**, arranged or suspended through the hole **3** or resting on an inner surface **5** of the hole **3**.
(27) Further, the second portion **12** has a second sub-portion **12***b* intended to form the rest **4** on the rear surface **2***b* of the wall **2**. This second sub-portion **12***b* is arranged preferably transversely with respect to the first sub-portion **12***a* and along the rear surface **2***b* of the wall **2**.
(28) The sub-portion **12***b* can have different geometric shapes so as to constitute different types of rest **4** like, for example, a point-like, linear or surface rest. Preferably, both the first portion **11**, comprising the first transverse element **11***c* and the second transverse element **11***d*, and the second portion **12**, comprising the first sub-portion **12***a* and the second sub-portion **12***b*, can be made into separable and reciprocally assemblable modules. This separation into modules can apply to one or more of the elements constituting the support body **10**.
(29) Alternatively, the different elements disclosed above that constitute the support body **10** can be made “in a single piece”.
(30) The fixing body **100** can be reversibly engaged with the first portion **11** and/or with the second portion **12** when the object-hanging device **1** is in a use configuration. The fixing body **100** exerts a first thrust action “A” and a second thrust action “B”, opposite the first thrust action “A”, respectively acting on the first portion **11**, and/or on the second portion **12**, and on the wall **2** so as to define stable anchoring of the support body **10** on the wall **2**. In the embodiment illustrated in the appended figures, the fixing body **100** is arranged along the frontal surface **2***a* of the wall **2**, engaging respectively with the support body **10**, and in particular with the first section **11** and/or with the second section **12**, and the frontal surface **2***a*.
(31) The fixing body **100** thus exerts thrust action “B” on the frontal surface **2***a* of the wall **2**. The rest **4** on the other hand helps couple the device **1** with the wall **2**, by means of a thrust action “C” on the rear surface **2***b* of the wall **2**. As this thrust action “C” opposes the action “B”, the device **1** performs a “gripping” action relative to the wall **2**.
(32) If the second transverse element **11***d* is present, the transverse element **11***d* exerts a further thrust action “D” on the frontal surface **2***a* of the wall **2** opposing the weight “P” of the external object engaged on the device **1**.
(33) Preferably, this transverse element is oriented to the floor where the support wall rests, exerting the thrust action “D” in an area on the frontal surface **2***a* of the wall arranged below the holes **3** receiving the second sub-portion **12***b* so as to provide support from below to the support body **10**.
(34) In the embodiment illustrated in the appended figures, the fixing body **100** is so constituted as to comprise at least one first seat **101** suitable for receiving at least one longitudinal portion of the first portion **11** of the support body **10**. This seat **101** can be so engaged as to engage the first portion **11** and/or the support body **100**.
(35) This engagement can be inside the fixing body **100** or, alternatively, outside, so as to solidly anchor the first portion **11** to the support body **100** by means of suitable connecting members and to avoid involuntary shifts of the first portion relative to the support body **100** and, consequently,

relative to the wall **2**.
(36) The fixing body **100** is also configured prevent rotation of the support body **10** relative to the wall **2**.
(37) With reference to the appended figures, the fixing body can comprise a main component **110**, suitable for exerting the first thrust action "A" and the second thrust action "B", and a support component **120**, reversibly connected to the main component **110** and suitable for preventing rotation of the support body **10**, in particular around the extension axis "S" of the support body **10**. The main component **110** can be used operationally autonomously or in conjunction with the support component **120**.
(38) In particular, in the embodiment of the appended figures, the main component **110** is defined preferably by a polyhedral geometry, preferably a prismatic geometry, even more preferably a six-face geometry arranged in a reciprocally orthogonal manner. For example, in the appended figures the main component **110** has a substantially cuboid conformation. In other embodiments that are not illustrated, the main component can have other geometries.
(39) In particular, the main component **110** comprises an upper face **110***a* and a lower face **110***b* arranged preferably parallel to the floor on which the support wall **2** is arranged, two side faces **110***c* and **110***d* preferably parallel, a frontal face **110***e* and a rear face **110***f* preferably parallel to the frontal surface **2***a* of the wall **2**. The rear face **110***f* is adapted to be arranged adhering to the wall **2**, and in particular to the frontal surface **2***a* so as to apply the thrust action "B" preferably in an extended manner against the frontal surface **2***a* of the wall **2**.
(40) The main component **110** defines the first seat **101**, adapted to receive the first portion **11** of the support body **10**, as a channel arranged along the lower face **110***b* of the main component **110**, compatible in form with the support body **10** and of the arrangement thereof relative to the frontal surface **2***a* of the wall **2**.
(41) Preferably, the main component **110** can be hollow internally so as to require, advantageously, less material for its manufacture and to be lighter. The support component **120** is made in polyhedral form, substantially with six reciprocally orthogonal sides, and more precisely as a plate, comprising an upper side **120***a* and a lower side **120***b* that are substantially more extensive than the remaining sides and are arranged parallel to the floor, two lateral sides **120***c* and **120***d*, a frontal side **120***e* and a rear side **120***f*.
(42) The support component **120** is couplable with the main component **110** by means of suitable connecting members, in particular having the upper face **120***a* reversibly connectable to the lower face **110***b* of the main component **110**.
(43) In this embodiment, the connecting members of the main component **110** comprise at least two hooks **111***a* and **111***b*, preferably arranged in the form of a "C" on the lower face **110**, and with respective facing cavities, and adapted to slidingly receive between them the support component **120**, which is provided in turn with two suitable sliding guides **121***a* and **121***b* arranged correspondingly along the lateral sides **120***e* and **120***f*. In particular, the support component **120** is inserted frontally, i.e. with the frontal side **120***e* facing the observer and the rear side **120***f* facing the frontal surface **2***a* of the wall relative to the main component **110**, preferably until the rear side **120***f* adheres to the frontal surface **2***a* of the wall **2**.
(44) Once the support component **120** is connected to the main component **110**, it exerts a fixing action by mechanical interference on the first portion **11** of the support body **10**, preventing the latter from being subjected to rotational stress around the axis "S".
(45) In the presence of the second transverse element **11***d*, the support component **120** defines a second seat **122** that is suitable for receiving the second transverse element **11***d*, to ensure simultaneously both translational and rotational anchoring of the support body **10** relative to the wall **2**. In particular, this second seat **122** is in a channel that is compatible in shape with the second transverse element **11***d* and arranged on the rear face **120***f* of the support component **120**. The arrangement enables the second transverse element **11***d* to be secured against both rotational and

translational stresses, being in further assistance to the first portion **11** immune to rotations in relation to the “S” axis.
(46) Preferably, and as for example shown in the appended figures, the support body **10** further comprises a third portion **13**, defined between the first portion **11** and the second portion **12**, defining a housing **14**, for or a coupling element with, the fixing body **100**.
(47) Preferably, the fixing body **100** is at least partially inserted into the housing **14**.
(48) Preferably, the housing **14** has a geometry that is such as to be compatible in form with the fixing body **100**, in other words so that the fixing body fills the housing **14** totally.
(49) Preferably, the first portion **11** and/or the second portion **12** and/or the third portion **13** are made in the form of thread-like elements.
(50) In the embodiment of the appended figures, the first portion **11**, the second portion **12** and the third portion **13** are made in the form of thread-like elements.
(51) Different embodiments can provide for each of these elements having different geometries, for example “I” or “L” or “V” or “C” or “U” or “M” geometries.
(52) In the embodiment of the appended figures, the second portion **12** is made “as one piece” with the third portion **13** so as to define a fork geometry.
(53) Preferably, the second portion **12** defines at least one tine of the fork geometry.
(54) In other words, the third portion **13** is substantially linear and arranged transversely to the second portion **12**.
(55) With reference to the appended figures, the second portion **12** defines two tines. For each tine, there is a corresponding sub-portion **12***a*, insertable singly into a corresponding hole **3** on the frontal surface **2***a* and a second sub-portion **12***b*, being like a natural extension of the first sub-portion **12***a* and forming a corresponding rest **4**.
(56) The housing **14** is preferably defined as a planar substantially rectangular shape, comprised between the third portion **13** and the frontal surface **2***a* of the wall **2** where the fixing body **100** is partially inserted from above, or is inserted from the lower surface **110***b*.
(57) In this arrangement, the thrust action “B” is exerted on a zone of the frontal surface **2***a* of the wall **2**. The rest **4** helps anchor the device **1** on the wall **2**, by means of a thrust action “C” on the rear surface **2***b* of the wall **2**.
(58) The second sub-portion **12***b* of the second portion **12** is oriented so as to obtain the rest **4** in an area arranged above the hole **3** traversed by the sub-portion **12***a*. The thrust “C” and the thrust “B” are symmetrical so that they form a grip around the wall **2**.
(59) By inserting the fixing body **100**, or possibly the main component **110** from above, the first seat **101** is made to rest on the first portion of the support body **11** and couple therewith.
(60) Preferably, the fixing body **100**, or possibly the main component **110**, comprises at least partially along the lateral perimeter, in other words at least along the frontal face **110***e* and/or the side faces **110***c* and **110***e*, a channel **112** suitable for receiving the third portion **13** of the support body **10**, when the fixing body **100** is inserted into the housing **14** so as to exert retaining by mechanical interference.
(61) Observing FIG. **6** or **7**, the fixing body is inserted inside the housing **14** by first resting the frontal face **110***e* and the fixing body on the third portion **13**, so that the third portion **13** engages with the channel **112**. Subsequently, by means of rotation, all the perimeter of the fixing body **100** is engaged inside the housing **14** and the main component is engaged by means of mechanical interference with the frontal surface **2***a* of the wall **2**.
(62) In the embodiment illustrated in the appended figures, the fixing body **100**, or possibly the main component **110**, exerts the thrust action “A” also at least partially on the third portion **13**.
(63) Further, the fixing body **100**, or possibly the main component **110**, exerts a lateral thrust action “E” on the third portion **13**, exerting anchoring that is free of possible lateral translations of the support body **10** and, consequently, of the device **1** relative to the wall **2**.
(64) In this embodiment, the third portion **13** and the first portion **11** are separate entities, which are

reversibly connectable by mechanical interference. The fixing body **100** is thus further configured to maintain the third portion **13** and the first portion **11** coupled, helping in this mechanical interference. Alternatively, the third portion **13** and the first portion **11** can be made "of one piece". In the circumstances in which the support body **10** provides for the use of transverse elements, the third portion **13** defines a housing **14** in which at least one of the transverse elements is at least partially inserted inside the housing **14**. In the embodiment of the appended figures, the second transverse element **11***d*, resting on the frontal surface **2***a* of the wall **2**, is inserted into the housing **14** defined by the third portion **13**. The second seat **122**, arranged on the support component **120**, is suitable for coupling through shape coupling with the second transverse element **11***d* to fix stably the transverse element **11***d* to the wall **2** and prevent rotating and/or translating against the wall **2**, and accordingly also rotating around the "S" of the support body **10**.
(65) With reference to FIGS. **10** and **15**, the support component **120** comprises a first slit **123***a* and a second slit **123***b* arranged opposite near the lateral sides **120***c* and **120***d.*
(66) This first and second slit define near lateral sides **120***c* and **120***d* respectively a first fin **124***a* and a second fin **124***b*, configured to be flexibly deformable with respect to the support component **120**.
(67) The fins are configured to engage with and disengage from the cavities defined respectively by the hooks **111***a* and **111***b.*
(68) The first and the second slit **123***a*, **123***b* are made with a course that is linear and parallel to the sliding guides **121***a* and **121***b.*
(69) Preferably, the first and the second slit **123***a* and **123***b* are through slits relative to the support component **120**.
(70) The support fins **124***a*, **124***b* engage by mechanical interference respectively with the cavities defined by the hooks **111***a* and **111***b*, enabling the support component **120** to be fixed in position relative to the main component **110**.
(71) The fins **124***a*, **124***b* are further configured to disengage respectively from the cavities defined by the hooks **111***a* and **111***b.*
(72) The lateral fins (**124***a*, **124***b*) are flexibly deformable relative to the support component (**120**) so as to engage, in use, the lateral fins (**124***a*, **124***b*) with the main component (**110**) and enable the lateral fins (**12***a*, **124***b*) to disengage from the main component (**110**).
(73) In other words, the first fin **124***a* and the second fin **124***b* are movable in relation to one another so as to reduce a distance therebetween so as to permit the fins to pass through the cavities defined by the hooks **111***a* and **111***b.*
(74) Preferably, said disengagement occurs by means of a manual compression action "F" of the fins **124***a*, **124***b* to lessen the mechanical interference between the fins **124***a*, **124***b* and the hooks **111***a* and **111***b.*
(75) Therefore, a pressure exerted by a user (i.e. the aforesaid manual compression action "F") enables this distance between the fins **124***a* and **124***b* to be so reduced that the support component **120** can be inserted or extracted to engage with or disengage from the main component **110**.
(76) In other words, during engagement of the support component **120** with the main component **110**, the first fin **124***a* and the second fin **124***b* are moved away from one another so as to define interference with the hooks **111***a* and **111***b* that prevents involuntary extraction thereof and only a manual compression action "F" thereof would allow an approach thereof to enable extraction thereof so as to disengage the support component **120** from the main component **110**.
(77) Advantageously, this configuration permits an engagement and a disengagement that are easily actuatable by the support component **120** relative to the main component **110** whereas it simultaneously stabilises in position the main component and, in turn, the support body **10**.
(78) The device **1** can comprise auxiliary elements adapted to improve the arrangement and the support of the external body with or on the support body **10**. Such auxiliary elements can comprise a rounded portion suitably arranged on the first portion **11**, and in particular on the first end **11***a* or

of the first transverse element **11***c*. This rounded portion comprises a cross section above the cross section of the first end **11***a* or of the first transverse element **11***c* so as to improve retaining of the external body and prevent an involuntary disengagement of the external body from the device **1**.
(79) This rounded portion can be integrated with the first portion **11** or be integratable as an accessory body **15**.
(80) Other alternative auxiliary elements that are integrated or are integratable with the first portion **11** of the support body **10** can comprise auxiliary bodies provided with coupling holes or auxiliary bodies provided with grippers.
(81) Further, the device **1** can also be employed by means of interposing a container that is clad on the first portion **11** and hanging on said wall **2**.
(82) Advantageously, the device **1** is capable of overcoming the drawbacks which have emerged from the prior art.
(83) Advantageously, using a device **1** comprising a support body **10** and a fixing body **100** that is reversibly engageable with the support body **10** permits reversible coupling with the support wall **2**, anyway ensuring that the device **1** is stably anchored on the wall and is not subjected to involuntary movements, whether in the event of impacts or of vibrations.
(84) Advantageously, the above device **1** is easy to install and uninstall.
(85) Further, the device **1** is realizable easily and economically.
(86) Further, the device **1** is easily reusable to support objects having different shapes, dimensions and functions.

Claims

1. An object-hanging device (**1**), intended to couple reversibly with a support wall (**2**) with a vertical extension and equipped with suitable through holes (**3**), wherein said device (**1**) comprises: a support body (**10**) comprising, a first portion (**11**) configured to make a rest element and/or a stop for an object; a second portion (**12**) configured to make a reversible connection between said support body (**10**) and said support wall (**2**); a fixing body (**100**) reversibly engageable or engaged with said first portion (**11**) and/or with said second portion (**12**) in a use configuration of said object-hanging device (**1**), wherein said second portion (**12**) is at least partially inserted inside at least one of said holes (**3**) so as to define the anchoring of said support body (**10**) on said wall (**2**); wherein said fixing body (**100**), in said use configuration, exerts a first thrust action (A) and a second thrust action (B), opposite said first thrust action (A), respectively acting on said first portion (**11**) and/or on said second portion (**12**), and on said wall (**2**) so as to define stable anchoring of said support body (**10**) on said wall (**2**); wherein said fixing body (**100**), in said use configuration, exerts a first thrust action (A) and a second thrust action (B), opposite said first thrust action (A), respectively acting on said first portion (**11**) and/or on said second portion (**12**), and on said wall (**2**) so as to define stable anchoring of said support body (**10**) on said wall (**2**), wherein said first portion (**11**) extends along a longitudinal axis(S) thereof and has a first end (**11***a*) and a second end (**11***b*) provided with respective transverse elements (**11***c* and **11***d*), said transverse elements (**11***c* and **11***d*), being opposite one another relative to said longitudinal axis (S), wherein said fixing body (**100**) comprises at least a first seat (**101**) suitable for receiving a longitudinal portion of said first portion (**11**) of the support body (**10**), wherein said fixing body (**100**) comprises a main component (**110**), suitable for exerting said first thrust action (A) and said second thrust action (B), and a support component (**120**) reversibly connected to said main component (**110**), and wherein said main component defines said first seat (**101**) and wherein said support component (**120**) defines a second seat (**122**) that is suitable for receiving one of said transverse elements (**11***d*).
2. The device (**1**) according to claim 1, wherein said fixing body (**100**), in said use configuration, is further configured to prevent rotation of said support body (**10**).

3. The device (**1**) according to claim 2, wherein said support component (**120**) is suitable for preventing rotation of said support body (**10**).
4. The device (**1**) according to claim 1, wherein said support body (**10**) comprises a third portion (**13**), defined between said first portion (**11**) and said second portion (**12**), defining a housing (**14**) for, or an element for coupling with, said fixing body (**100**).
5. The device (**1**) according to claim 4, wherein said second portion (**12**) is made in a single piece with said third portion (**13**) so as to define a fork geometry, wherein at least said second portion (**12**) defines at least one tine of said fork geometry.
6. The device (**1**) according to claim 4, wherein said third portion (**13**) defines a housing (**14**) for said fixing body (**100**) and wherein said fixing body (**100**) comprises at least partially along a perimeter a channel (**112**) adapted to receive said third portion (**13**) of said support body (**10**).
7. The device (**1**) according to claim 4, wherein said first portion (**11**) and/or said second portion (**12**) and/or said third portion (**13**) are made in the form of threadlike elements having an “I” or an “L” or a “V” or a “C” or a “U” or an “M”-shaped geometry.
8. The device (**1**) according to claim 4, wherein said third portion (**13**) and said first portion (**11**) are reversibly connectable by mechanical interference, and wherein said fixing body (**100**) is further configured to maintain said mechanical interference between the third portion (**13**) and the first portion (**11**).
9. The device according to claim 4, wherein said third portion (**13**) defines a housing (**14**) and wherein at least one of said transverse elements (**11***d*) is at least partially inserted inside said housing (**100**).
10. The device according to claim 3, wherein the support component is couplable with said main component (**110**) by suitable connecting members, said connecting members comprising: at least two “C”-shaped hooks (**111***a* and **111***b*), arranged on the main component (**110**) and with respective facing cavities; two sliding guides (**121***a* and **121***b*), arranged correspondingly along two lateral sides (**120***c* and **120***d*) of said support component (**120**); wherein said hooks (**111***a* and **111***b*) are adapted to slidingly receive between them the support component (**120**) and realize a connection therewith by mechanical interference.
11. The device according to claim 10, wherein the support component (**120**) comprises a first slit (**123***a*) and a second slit (**123***b*) defining a first lateral fin (**124***a*) and a second lateral fin (**124***b*) that are arranged respectively near said lateral sides (**120***e* and **120***d*), said lateral fins (**124***a*, **124***b*) being flexibly deformable relative to said support component (**120**) so as to realise, in use, an engagement of the lateral fins (**124***a*, **124***b*) with the main component (**110**) and permit disengagement of the lateral fins (**124***a*, **124***b*) from the main component (**110**).